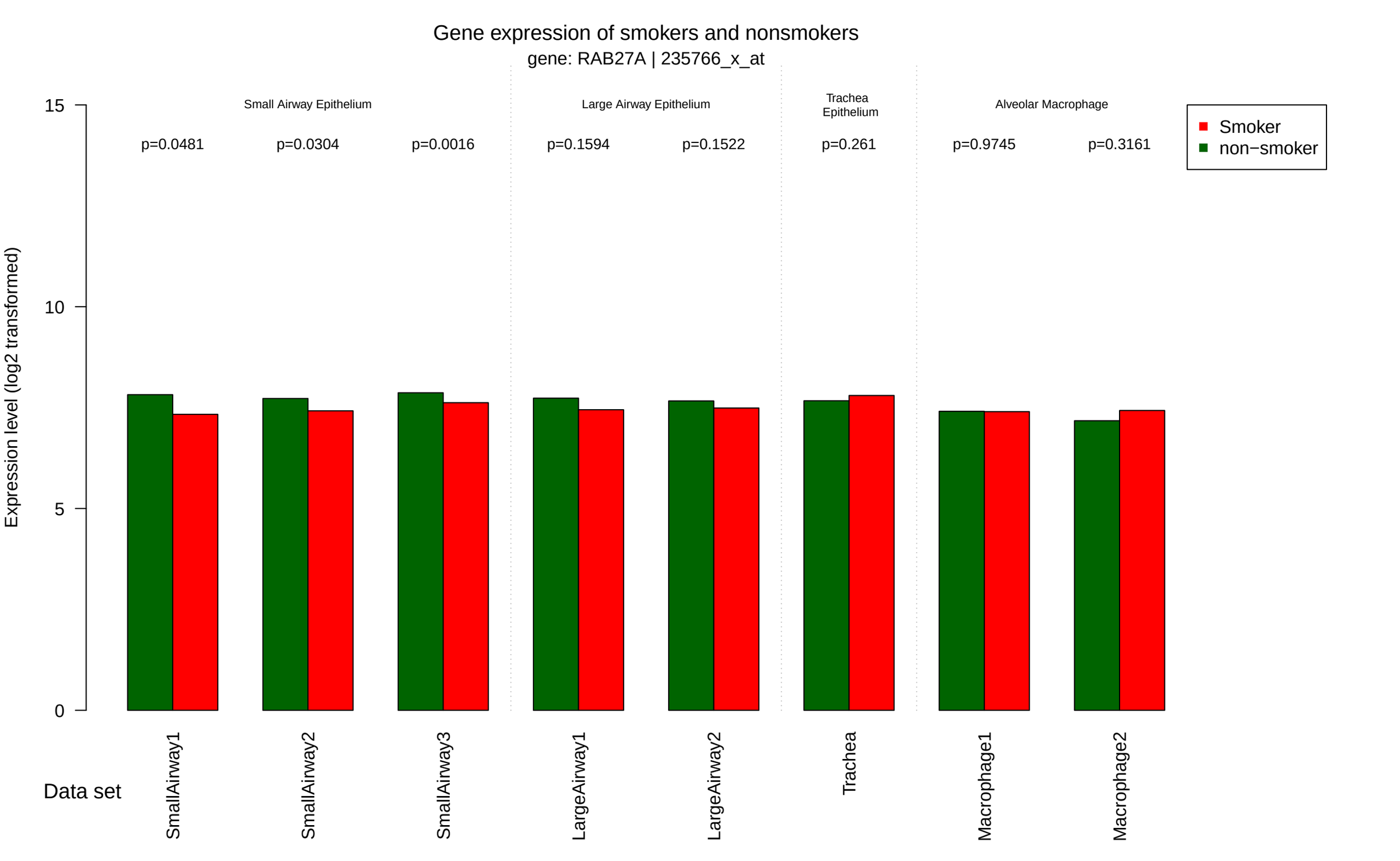
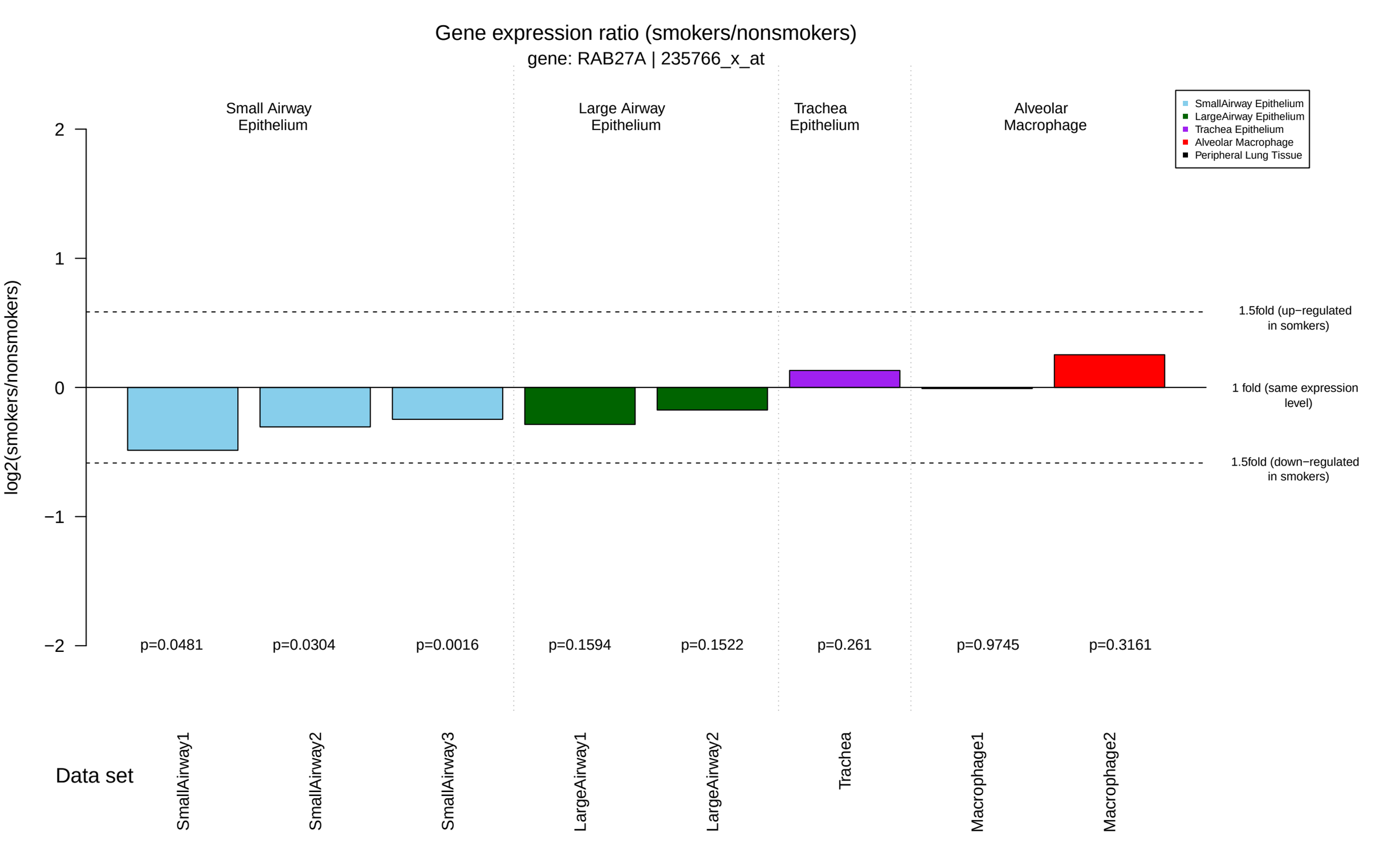
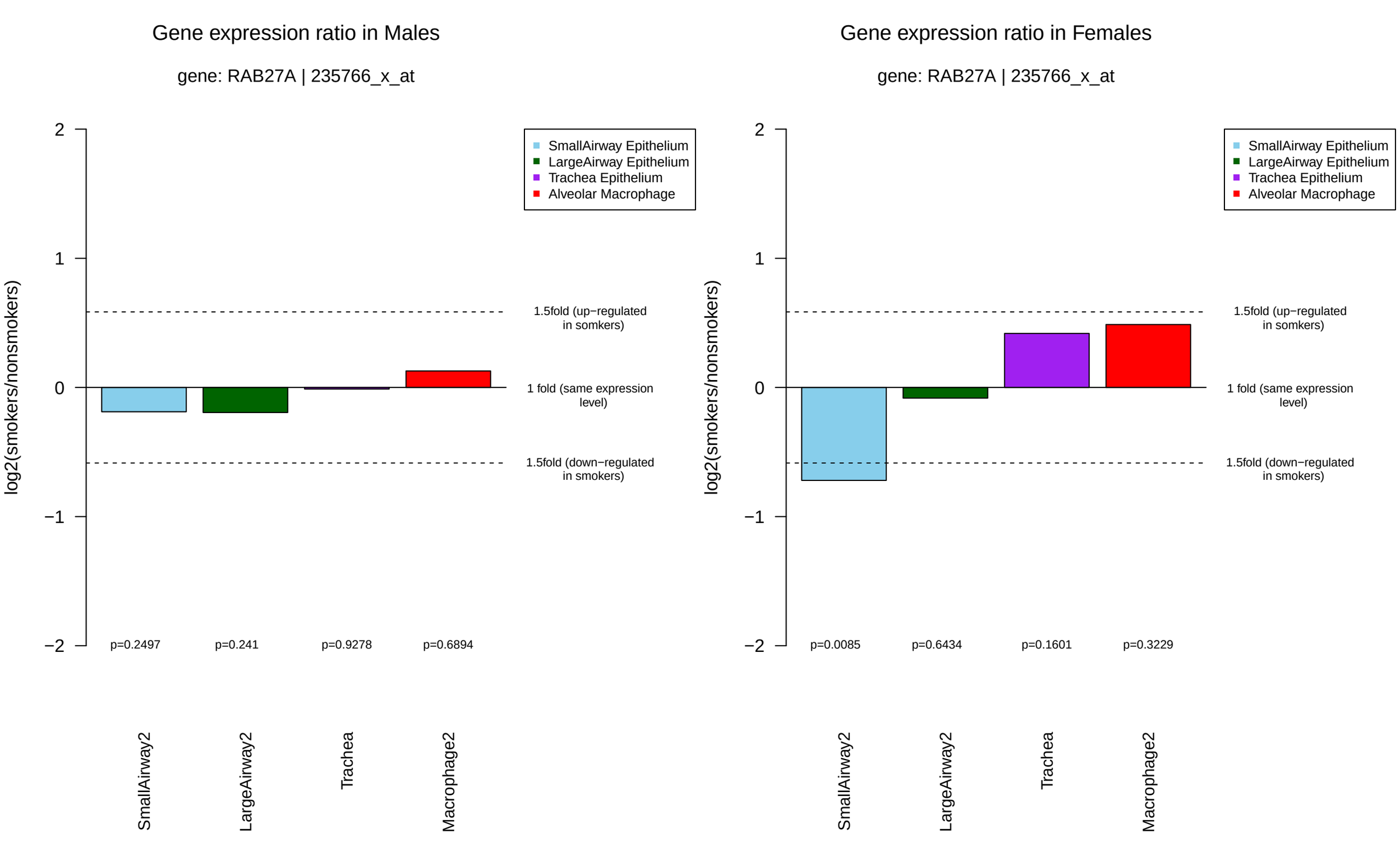
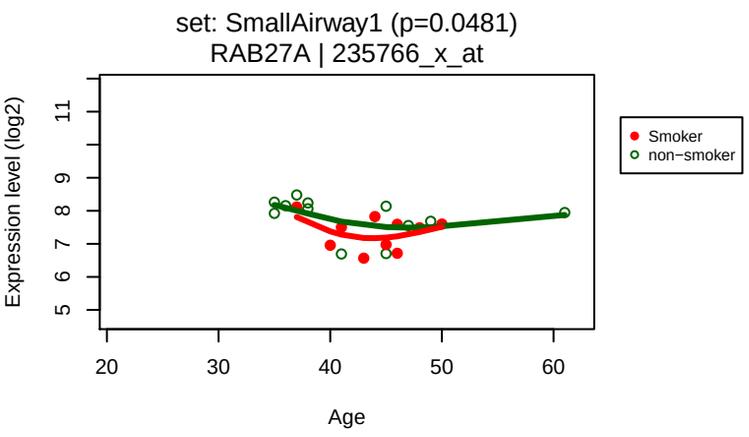
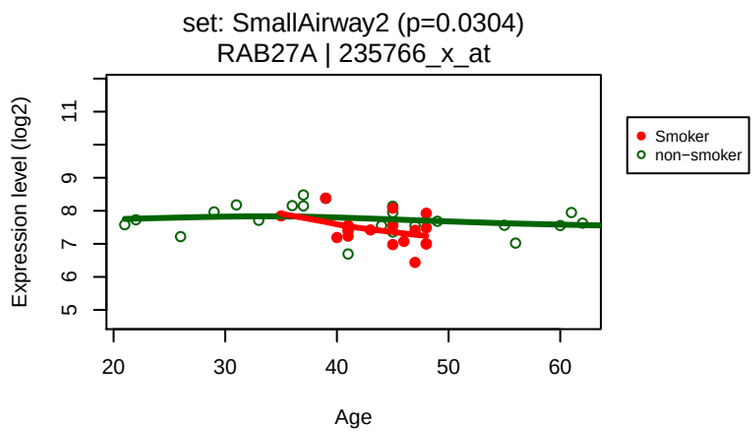
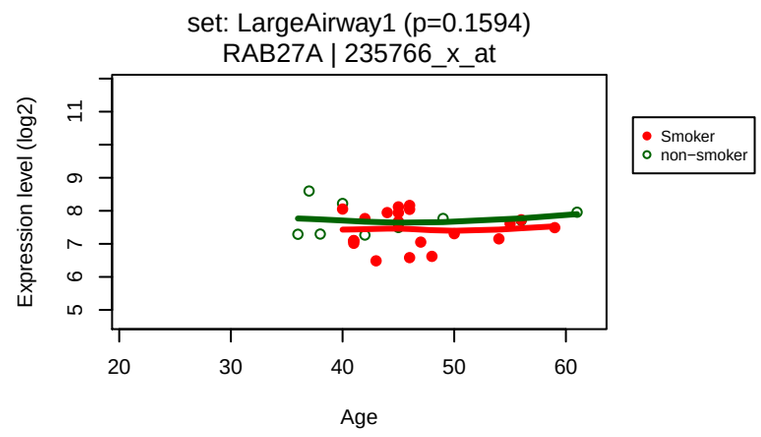

Gene expression of smokers and nonsmokers
gene: RAB27A | 235766_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0481
p=0.0304
p=0.0016
p=0.1594
p=0.1522
p=0.261
p=0.9745
p=0.3161
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

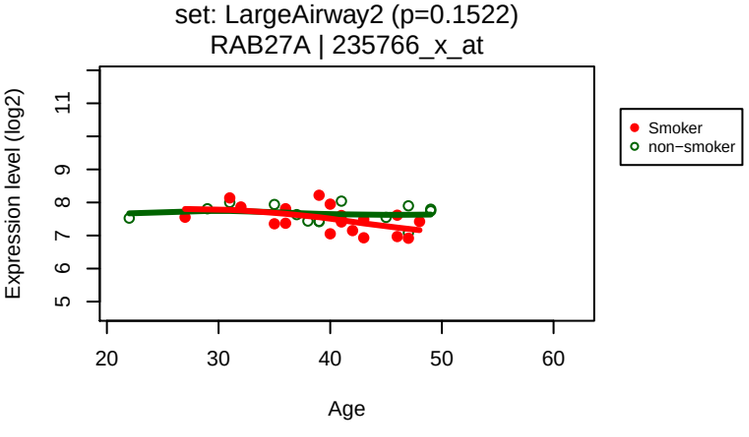

Gene expression ratio (smokers/nonsmokers)
gene: RAB27A | 235766_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0481
p=0.0304
p=0.0016
p=0.1594
p=0.1522
p=0.261
p=0.9745
p=0.3161
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

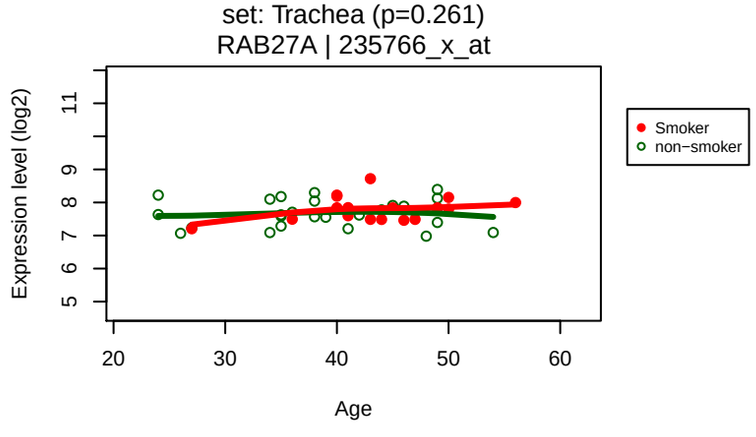

Gene expression ratio in Males
gene: RAB27A | 235766_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2497
p=0.241
p=0.9278
p=0.6894
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB27A | 235766_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0085
p=0.6434
p=0.1601
p=0.3229
SmallAirway2
LargeAirway2
Trachea
Macrophage2

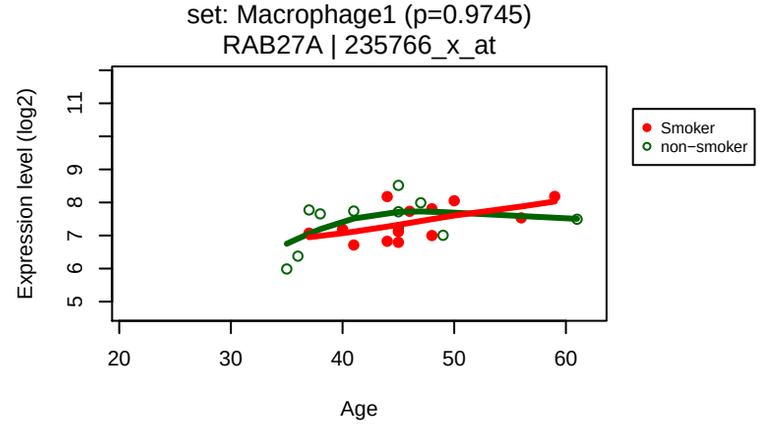

set: SmallAirway1 (p=0.0481)
RAB27A | 235766_x_at
Expression level (log2)
Age
Smoker
non−smoker

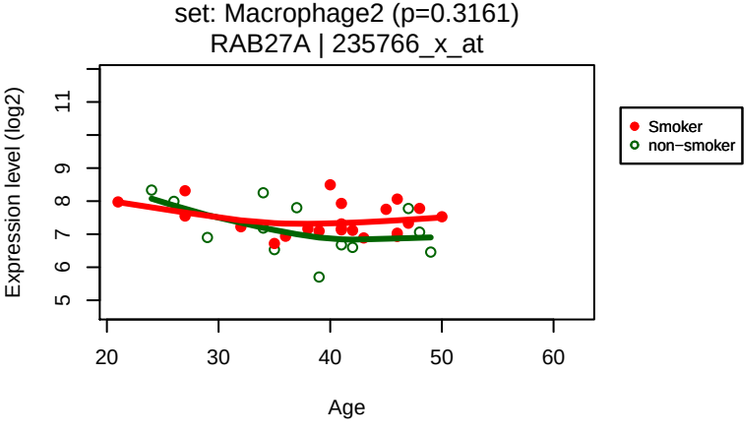

set: SmallAirway2 (p=0.0304)
RAB27A | 235766_x_at
Expression level (log2)
Age
Smoker
non−smoker

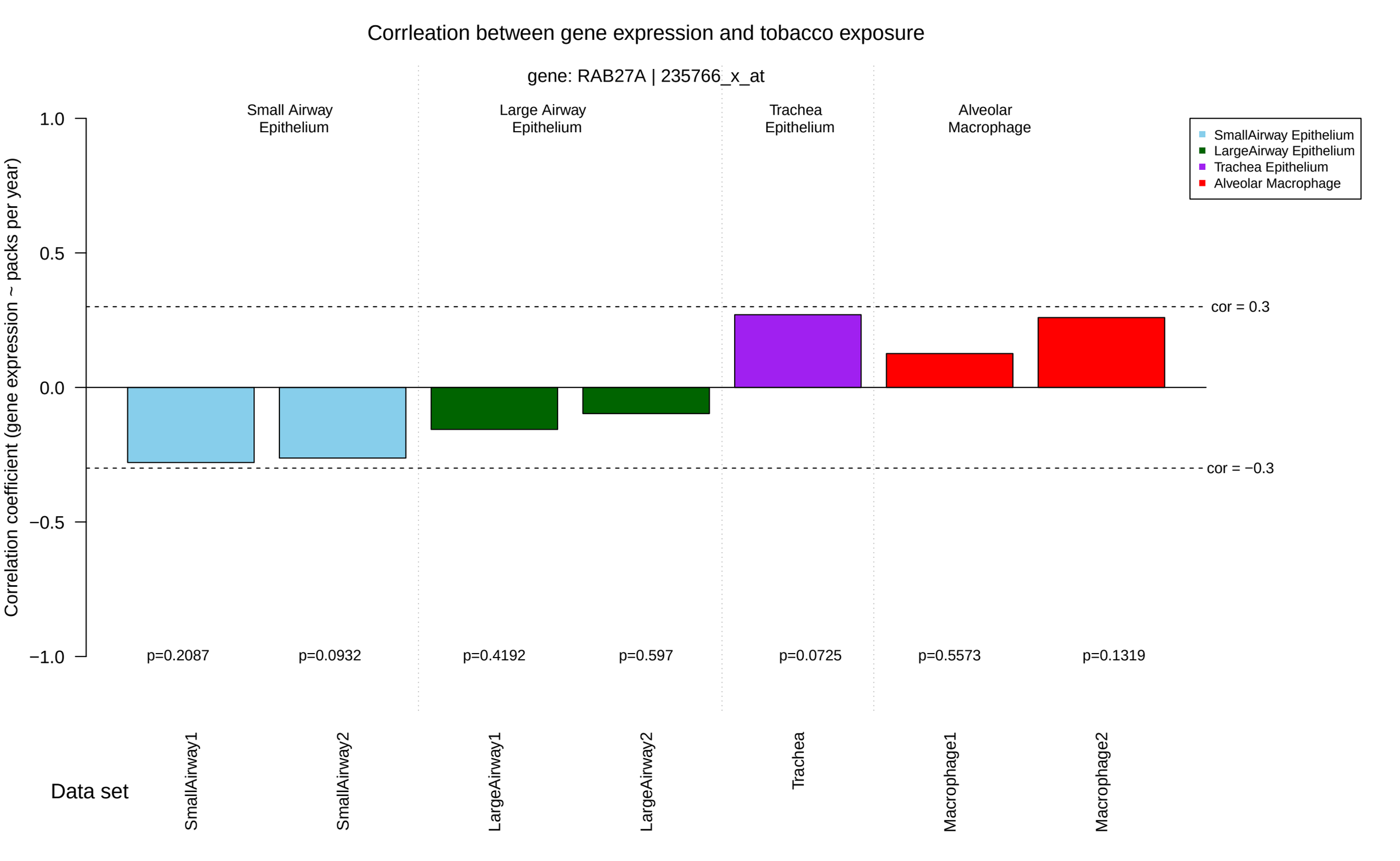

set: LargeAirway1 (p=0.1594)
RAB27A | 235766_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1522)
RAB27A | 235766_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.261)
RAB27A | 235766_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9745)
RAB27A | 235766_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3161)
RAB27A | 235766_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB27A | 235766_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2087
p=0.0932
p=0.4192
p=0.597
p=0.0725
p=0.5573
p=0.1319
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2